



 FACULTÉ
D'INFORMATIQUE

CHAPTER 3

RELATIONAL ALGEBRA

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Content

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What is the Relational Algebra?

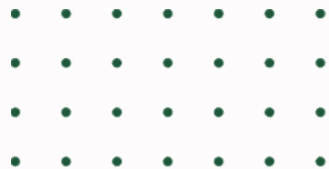
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Relational Algebra operators:

- Selection
- Projection
- Union, Intersection, Difference
- Cartesian Product
- Join
- Division

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Exercise



Introduction

- In last chapters, we studied how to design a relational database and how to implement it.
- In this chapter, we study how to query a relational database through relational algebra operators.

What is the Relational Algebra?

- Relational algebra is
 - a collection of algebraic operations each acting on one or two relations and producing one relation as result, and
 - a language for combining those operations
- Relational algebra is essential for the theoretical foundation of SQL (Structured Query Language) and plays a significant role in query optimization within DBMS.

Relational Algebra operators

- **Selection**
- **Projection**
- **Union, Intersection, Difference**
- **Cartesian Product**
- **Join**
- **Division**

Relational Algebra operators

Course Example

Student

SID	Name	Major
S1	Alice	Computer Science
S2	Bob	Mathematics
S3	Carol	Computer Science

Course

CID	Title	Credits
C1	Databases	3
C2	AI	4

Enrolled

SID	CID	Grade
S1	C1	A
S1	C2	B
S2	C2	A

Relational Algebra operators

Selection (or Restriction) (σ)

Definition:

The selection operation is used to retrieve rows from a relation that satisfies a given condition.

Syntax:

$$\sigma_{\text{condition}}(\text{Relation})$$

Example:

Get students majoring in Computer Science.

$$\sigma_{\text{Major}=\text{'Computer Science'}}(\text{Student})$$

SID	Name	Major
S1	Alice	Computer Science
S3	Carol	Computer Science

Relational Algebra operators

Projection (π)

Definition:

The projection operation is used to retrieve specific columns from a relation (removes duplicates).

Syntax:

$$\pi_{(\text{column1, column2, ...})}(\text{Relation})$$

Example:

Get all student names.

$$\pi_{\text{Name}}(\text{Student})$$

Name
Alice
Bob
Carol

Get the student's major

$$\pi_{\text{Major}}(\text{Student})$$

Major
Computer Science
Mathematics

removes duplicates

Relational Algebra operators

Union ($\cup$)

Definition: The union operation combines the results of two relations, eliminating duplicates. It requires both relations to have the same number of attributes with matching domains.

Syntax:

Relation1 $\cup$ Relation2

Example:

Get students majoring in Computer science or Mathematics

$(\sigma (\text{Major}=\text{'Computer Science'})(\text{Student})) \cup (\sigma (\text{Major}=\text{'Mathematics'})(\text{Student}))$

SID	Name	Major
S1	Alice	Computer Science
S3	Carol	Computer Science

$\cup$

SID	Name	Major
S2	Bob	Mathematics

=

SID	Name	Major
S1	Alice	Computer Science
S2	Bob	Mathematics
S3	Carol	Computer Science

Relational Algebra operators

Intersection ($\cap$)

Definition: The intersection operation returns the rows that are common to both relations. It also requires the same structure for both relations.

Syntax:

Relation1 $\cap$ Relation2

Example:

SID	Name	Major
S1	Alice	Computer Science
S3	Carol	Computer Science

$\cap$

SID	Name	Major
S1	Alice	Computer Science
S2	Bob	Mathematics
S4	Carlos	Computer Science

=

SID	Name	Major
S1	Alice	Computer Science

Relational Algebra operators

Difference (-)

Definition:

Gives tuples in one relation but **not** the other.

Syntax:

Relation1 - Relation2

Example:

Students not enrolled in any course.

$\pi_{SID}(\text{Student}) - \pi_{SID}(\text{Enrolled})$

SID
S3

Carol (S3) is not enrolled.

Relational Algebra operators

Cartesian Product ($\times$)

Definition:

Combines every row from one relation with every row from another.

Syntax:

Relation1 $\times$ Relation2

Example:

Combine all students with all courses

Student $\times$ Course

SID	Name	Major	CID	Title	Credits
S1	Alice	Computer Science	C1	Databases	3
S1	Alice	Computer Science	C2	AI	4
S2	Bob	Mathematics	C1	Databases	3
S2	Bob	Mathematics	C2	AI	4
S3	Carol	Computer Science	C1	Databases	3
S3	Carol	Computer Science	C2	AI	4

Relational Algebra operators

Join

Definition:

Join operations are used to combine data from two relations based on some condition.

Types:

- Theta join
- Natural join
- Outer join

Relational Algebra operators

Natural Join

Definition: A natural join automatically joins two relations by matching columns with the same names and combines the rows where these columns have equal values.

Syntax:

Relation1 $\bowtie$ Relation2

Example: Get students with their enrolled courses.

(Student $\bowtie$ Enrolled) $\bowtie$ Course

Relational Algebra operators

Theta Join

Definition: A theta join combines rows from two relations based on a condition (theta) that can be any comparison operator (e.g., =, <, >, <=, >=, ≠).

Syntax:

Relation1 $\bowtie_{\theta}$ Relation2

Example: Find the students who are enrolled in courses with more than 3 credits.

(Student $\bowtie$ Enrolled) $\bowtie_{(\text{credit} > 3)}$ Course

Relational Algebra operators

Outer Join

Outer joins return rows that do not have a match in the other relation. The three types of outer joins are:

- **Left join** ($\bowtie$): Returns all rows from the left relation and the matching rows from the right relation (nulls for unmatched rows in the right relation).
- **Right join** ($\bowtie$): Returns all rows from the right relation and the matching rows from the left relation (nulls for unmatched rows in the left relation).
- **Full join** ($\bowtie$): Returns all rows when there is a match in either left or right relation (nulls for unmatched rows from both relations).

Relational Algebra operators

Division

Definition: The division operation is used to find tuples in one relation that are related to all tuples in another relation. It is typically used for "all" queries.

Syntax:

Relation1 $\div$ Relation2

Example:

Students who took **all** courses.

$\pi_{SID, CID}(\text{Enrolled}) \div \pi_{CID}(\text{Course})$

R		S	$T = R \div S$
A	B	B	A
a1	b1	b1	a1
a1	b2	b2	
a1	b3	b3	
a1	b4		
a2	b1		
a2	b3		
a3	b2		
a3	b3		
a3	b4		
a4	b1		
a4	b2		
a4	b3		

Relational Algebra operators

Combining Algebraic Operations

We can combine different operation to handle complex queries.

Example: Find the names of Computer Science students who have not enrolled in any course with more than 3 credits.

This involves:

- Selection:** Filter CS students.
- Join:** Match students with the courses they are enrolled in.
- Selection:** Pick courses with > 3 credits.
- Set difference:** Remove students who took any such course.
- Projection:** Return names only.

Final query: $\pi_{\text{Name}}((\pi_{\text{SID}}(\sigma_{\text{Major}=\text{'Computer Science'}}(\text{Student})) - \pi_{\text{SID}}(\sigma_{\text{Credits}>3}(\text{Student} \bowtie \text{Enrolled} \bowtie \text{Course}))) \bowtie \text{Student})$

Relational Algebra operators

Summary

Operator	Symbol	Use Case
Selection	σ	Filter rows
Projection	π	Select columns
Cartesian	$\times$	Combine all pairs
Join	$\bowtie$	Combine related rows
Union	$\cup$	Combine sets
Difference	$-$	Find what's only in one relation
Intersection	$\cap$	Common tuples
Division	$\div$	"All" queries

Relational Algebra operators

Exercise

Library System:

Book(BID, Title, Author)

Member(MID, Name, City)

Borrowed(MID, BID, BorrowDate)

- Find the title of books written by 'Kateb Yacine'
- Find the names of members who have borrowed books written by 'Kateb Yacine'
- Find the names of members who have borrowed books written by 'Kateb Yacine' or by 'Mouloud Feraoun'
- Find the names of members who have borrowed **all** books written by 'Kateb Yacine'



The END